

National Textile University

Department of Computer Science

Subject:

Dev Ops

Submitted to:

Mr Rehan Sarwar

Submitted by:

Abdul Rehman

Reg number:

23-NTU-CS-1122

Lab no. :

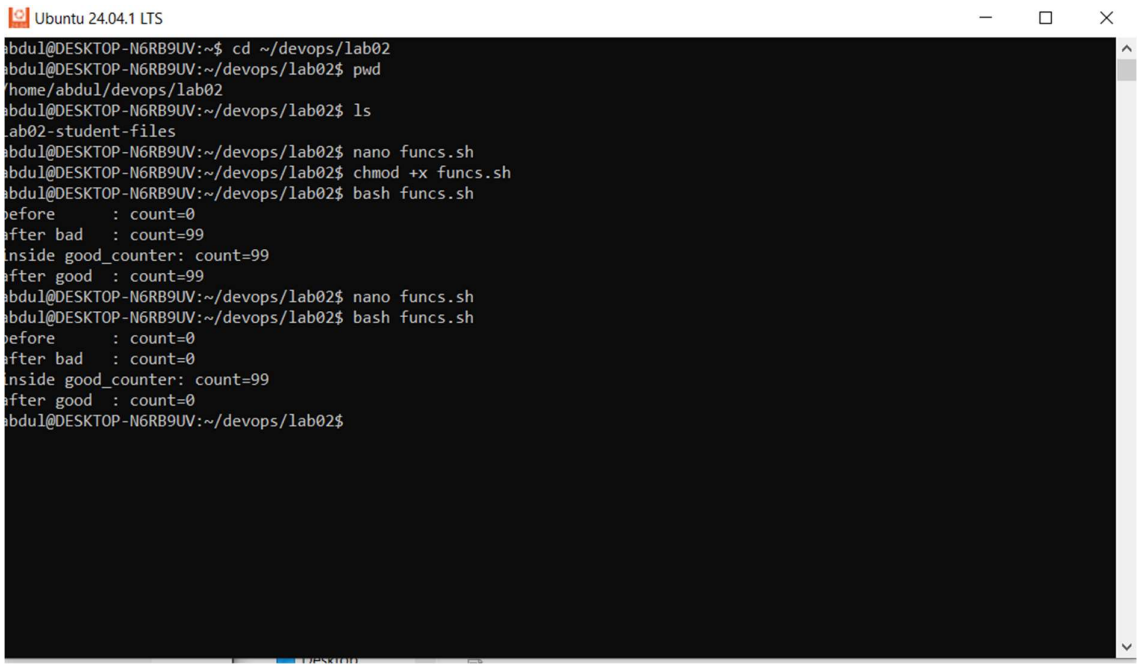
2

Semester:

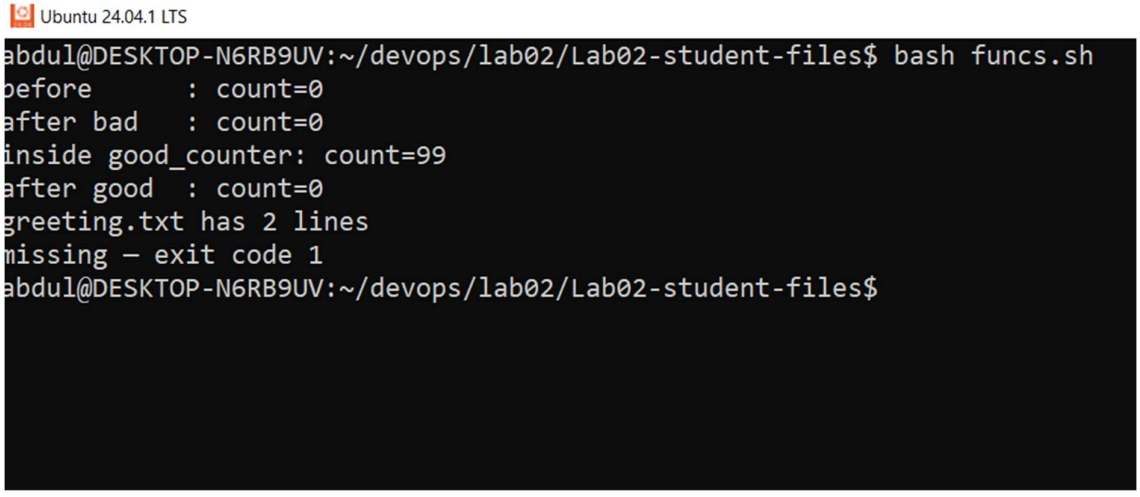
7th(A)

Task 1

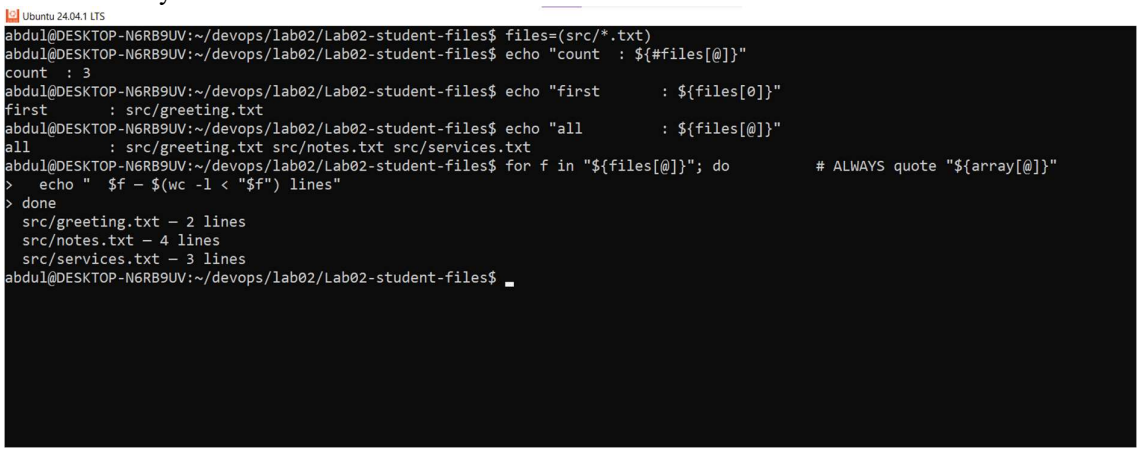
1(a) and (b)



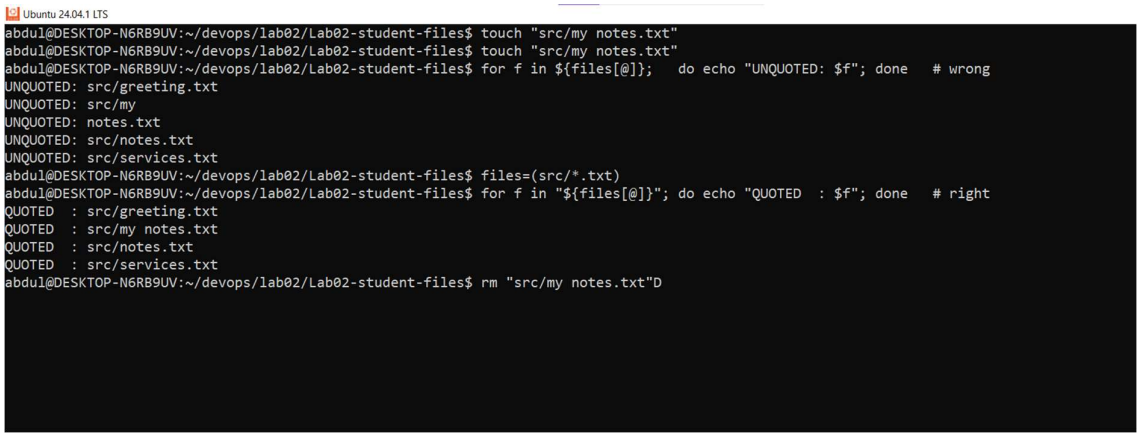
Task 1.2 — Return values exit codes



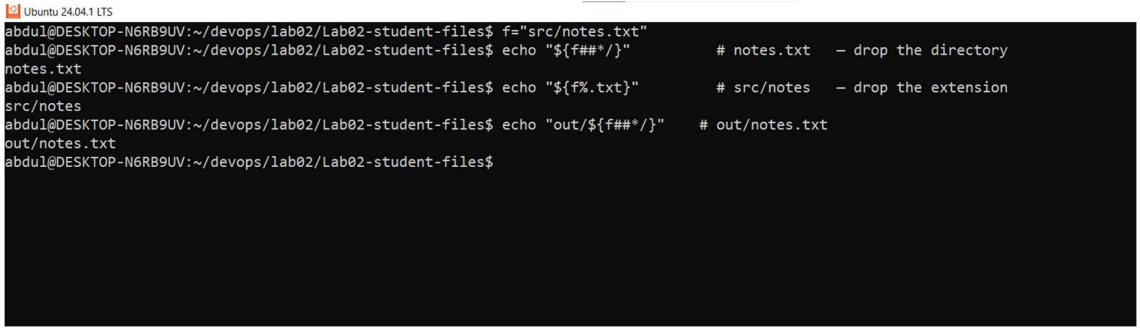
Task 1.3: Arrays



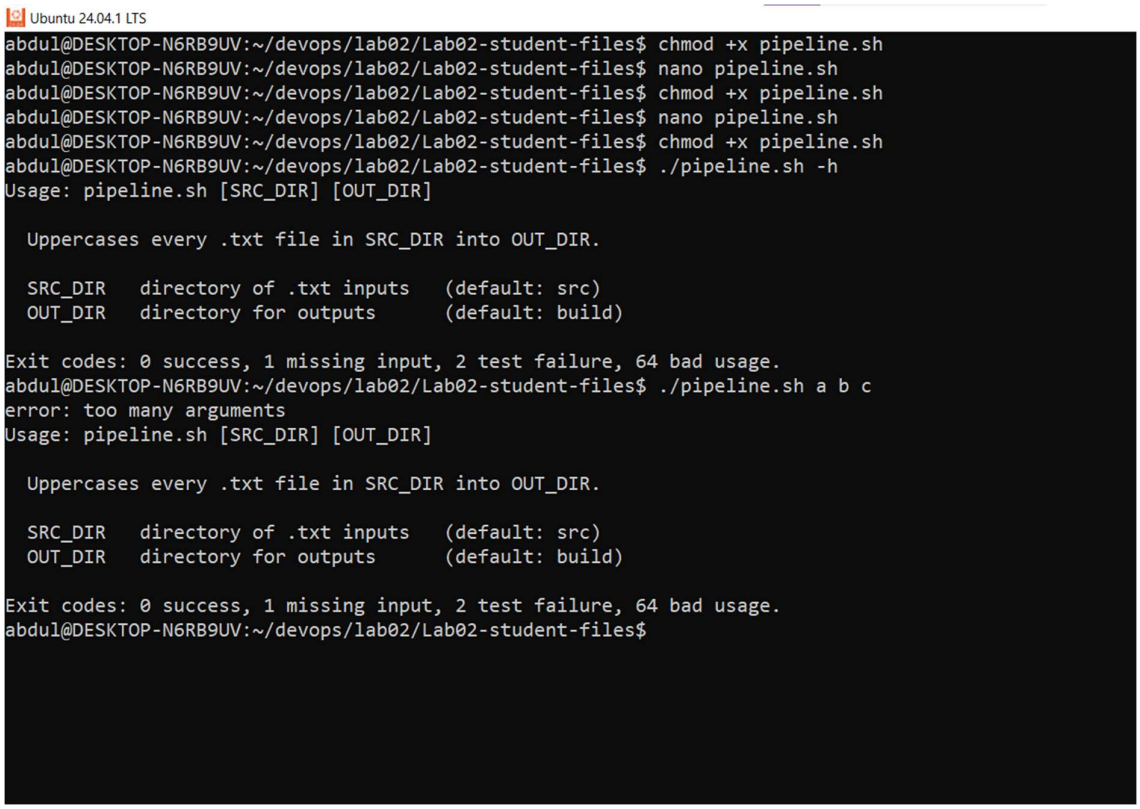
Quoted vs Unquoted

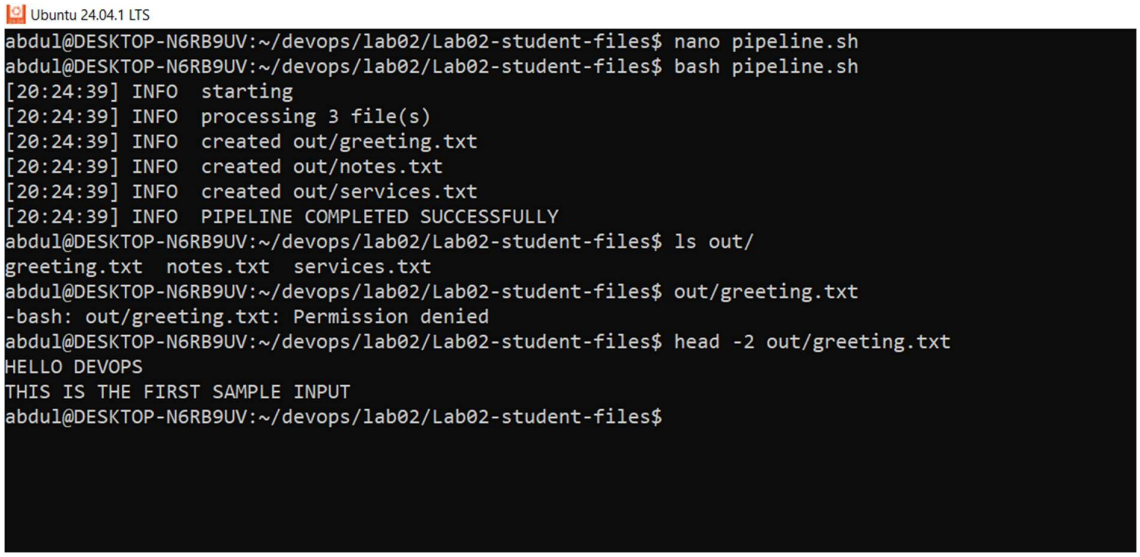


Task 1.4 — Parameter Expansion



Task 2, Step 1: pipeline.sh


Ubuntu 24.04.1 LTS
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ chmod +x pipeline.sh
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ nano pipeline.sh
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ chmod +x pipeline.sh
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ nano pipeline.sh
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ chmod +x pipeline.sh
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ ./pipeline.sh -h
Usage: pipeline.sh [SRC_DIR] [OUT_DIR]
Uppercases every .txt file in SRC_DIR into OUT_DIR.
SRC_DIR directory of .txt inputs (default: src)
OUT_DIR directory for outputs (default: build)
Exit codes: 0 success, 1 missing input, 2 test failure, 64 bad usage.
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ ./pipeline.sh a b c
error: too many arguments
Usage: pipeline.sh [SRC_DIR] [OUT_DIR]
Uppercases every .txt file in SRC_DIR into OUT_DIR.
SRC_DIR directory of .txt inputs (default: src)
OUT_DIR directory for outputs (default: build)
Exit codes: 0 success, 1 missing input, 2 test failure, 64 bad usage.
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$


Ubuntu 24.04.1 LTS
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ nano pipeline.sh
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ bash pipeline.sh
[20:24:39] INFO starting
[20:24:39] INFO processing 3 file(s)
[20:24:39] INFO created out/greeting.txt
[20:24:39] INFO created out/notes.txt
[20:24:39] INFO created out/services.txt
[20:24:39] INFO PIPELINE COMPLETED SUCCESSFULLY
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ ls out/
greeting.txt notes.txt services.txt
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ out/greeting.txt
-bash: out/greeting.txt: Permission denied
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ head -2 out/greeting.txt
HELLO DEVOPS
THIS IS THE FIRST SAMPLE INPUT
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$

Task 3.1

Ubuntu 24.04.1 LTS

```
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ cat -A Makefile
hello:$
echo "Hello from make"$
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$
```

Task 3.2:

Ubuntu 24.04.1 LTS

```
mkdir -p out
tr '[:lower:]' '[:upper:]' < src/greeting.txt > out/greeting.txt
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ make out/greeting.txt
make: 'out/greeting.txt' is up to date.
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ touch src/greeting.txt
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ make out/greeting.txt
mkdir -p out
tr '[:lower:]' '[:upper:]' < src/greeting.txt > out/greeting.txt
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$
```

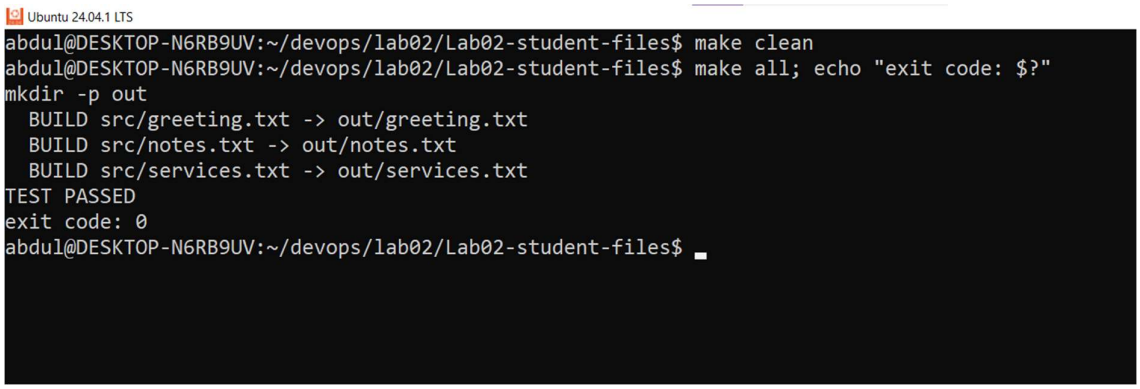
Task 3.3

Ubuntu 24.04.1 LTS

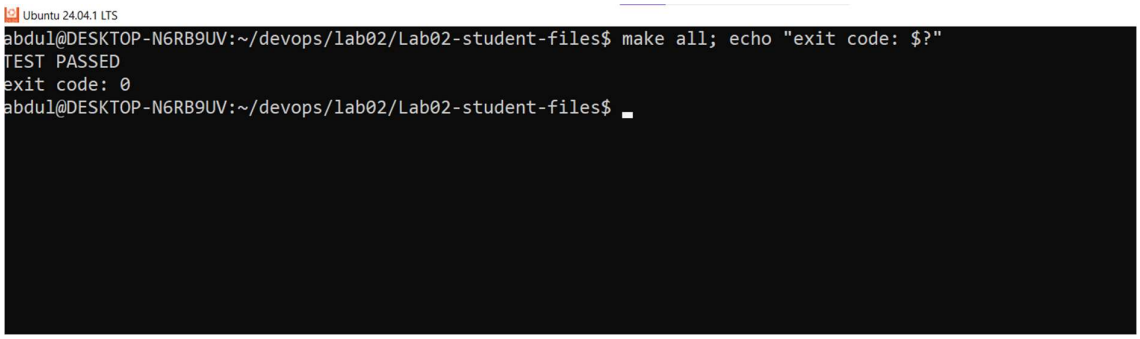
```
tr '[:lower:]' '[:upper:]' < src/greeting.txt > out/greeting.txt
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ make out/greeting.txt
make: 'out/greeting.txt' is up to date.
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ touch src/greeting.txt
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ make out/greeting.txt
mkdir -p out
tr '[:lower:]' '[:upper:]' < src/greeting.txt > out/greeting.txt
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ nano Makefile
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ nano Makefile
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ make clean
rm -rf out
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ ls
Makefile  Makefile.starter  README.txt  build  build.v1.sh  funcs.sh  pipeline-task1-backup.sh  pipeline.sh  src
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ touch clean
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ make clean
rm -rf out
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$
```

Task 3.4:

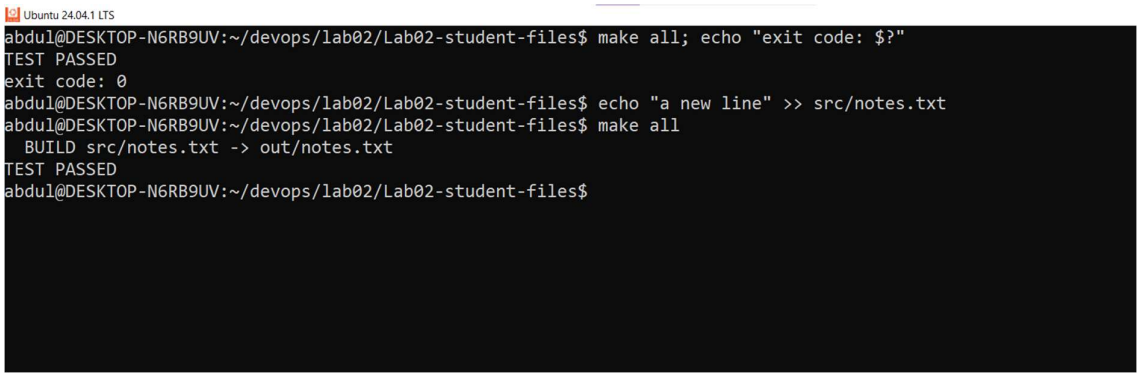
Task 4.1:



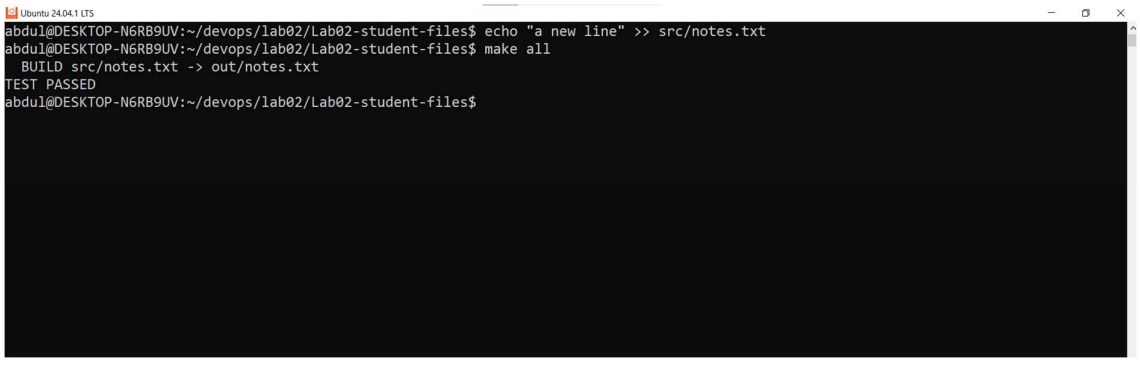
Task 4.2



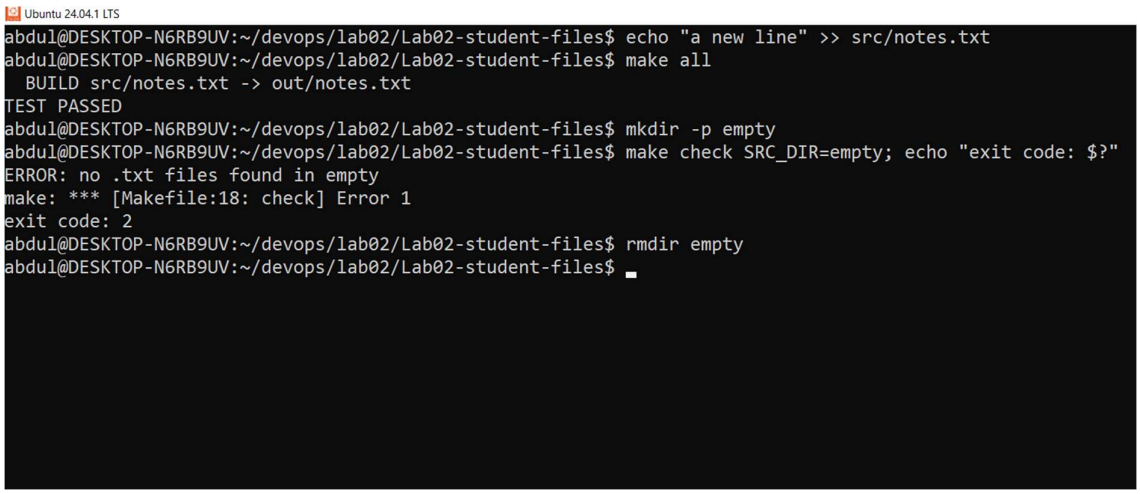
4.3



4.4



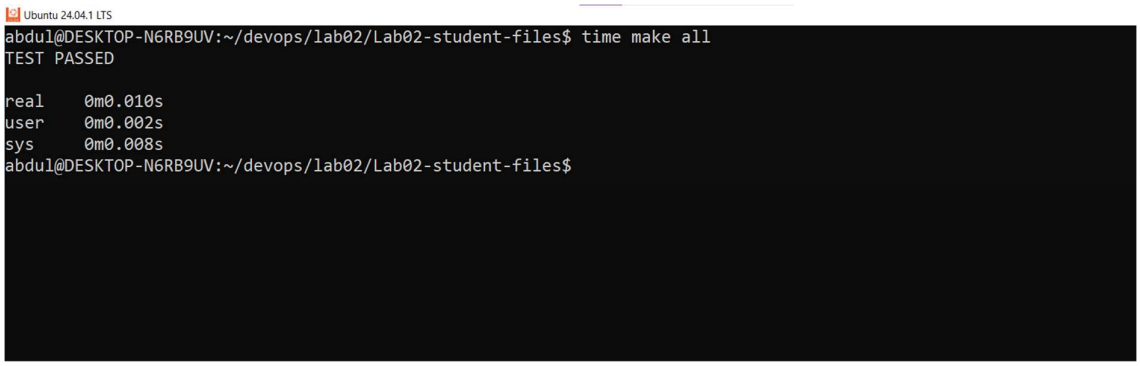
4.5


Ubuntu 24.04.1 LTS
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ echo "a new line" >> src/notes.txt
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ make all
BUILD src/notes.txt -> out/notes.txt
TEST PASSED
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ mkdir -p empty
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ make check SRC_DIR=empty; echo "exit code: $?"
ERROR: no .txt files found in empty
make: *** [Makefile:18: check] Error 1
exit code: 2
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ rmdir empty
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$

5.1


Ubuntu 24.04.1 LTS
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ make clean
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ time make all
mkdir -p out
BUILD src/greeting.txt -> out/greeting.txt
BUILD src/notes.txt -> out/notes.txt
BUILD src/services.txt -> out/services.txt
TEST PASSED
real 0m0.025s
user 0m0.011s
sys 0m0.013s
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$

5.2


Ubuntu 24.04.1 LTS
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ time make all
TEST PASSED
real 0m0.010s
user 0m0.002s
sys 0m0.008s
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$

5.3

```
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ touch src/notes.txt
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ time make all
  BUILD src/notes.txt -> out/notes.txt
TEST PASSED

real    0m0.009s
user    0m0.006s
sys     0m0.003s
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$
```

5.4

```
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$ time bash pipeline.sh
[21:20:31] INFO  starting
[21:20:31] INFO  processing 3 file(s)
[21:20:31] INFO  created out/greeting.txt
[21:20:31] INFO  created out/notes.txt
[21:20:31] INFO  created out/services.txt
[21:20:31] INFO  PIPELINE COMPLETED SUCCESSFULLY

real    0m0.056s
user    0m0.023s
sys     0m0.018s
abdul@DESKTOP-N6RB9UV:~/devops/lab02/Lab02-student-files$
```

Comparison

Case	real time	Files built	Notes
pipeline.sh (no dependency tracking)	3s	3	Rebuilds everything, always
make all — cold	3s	3	Everything is built from scratch
make all — warm	0s	0	Nothing needs rebuilding
make all — one file changed	1s	1	Only changed file is rebuilt

Reflection:

Problem/Reflection:

During testing, I used an unset variable $TMP. Because the script uses set -u, Bash stopped the script and showed an error. This helped me understand that set -u detects variables that are not defined and prevents the script from continuing with unexpected values. I then removed the test line and ran the script again successfully.